

Davide Ottaviani

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Education

Sapienza University of Rome, MS in Artificial Intelligence and Robotics Oct 2025 - Present

- **Coursework:** Advanced Artificial Intelligence, Machine Learning, Robotics, Computer Vision, Natural Language Processing, Reinforcement Learning, Controls, Autonomous System

Sapienza University of Rome, BS in Computer and Control Engineering Sept 2022 - Oct 2025

- Final Grade: 110/110
- **Coursework:** Computer Architecture, Algorithms and Complexity, Computational Theory, Fundamentals of Artificial Intelligence, Data Structures, Controls and Programming

Projects

Joint Detection of AI-Generated Images & Post-Processing github.com/cv_project_2526

- Computer Vision course project: a dual-task pipeline detecting AI-generated images and identifying their post-processing (original / transmitted / re-digitized), built on a frozen DINOv3 backbone with an FFT frequency branch and LoRA adapters.
- Compared Baseline, Joint, and LoRA training regimes across 6 configurations; with full ablations on loss weighting and shared-trunk bottleneck size.
- Tools Used: PyTorch, DINOv3 (ViT-B/16), LoRA/PEFT, Transformers

Multilingual Semantic Search & Retrieval

- Natural Language Processing course project: an information retrieval system built on a bi-encoder architecture, comparing fine-tuning strategies (Sentence Pairs vs. Triplets) with a cross-encoder reranking stage.
- Fine-tuned DistilBERT and all-MiniLM with MultipleNegativesRankingLoss; best pipeline (Triplets + reranking).
- Tools Used: Python, SentenceTransformers, PyTorch

Interactive CC8 - Procedural Mars Rover 3D Game

live demo

- Browser-based 3D driving game built from scratch with Three.js: a rover crosses a procedurally generated Martian desert with a physically animated rocker-bogie suspension, a day/night cycle, and a dependency-driven mission system.
- Implemented infinite procedural terrain (simplex-noise fBm with seamless chunk streaming) and live-synthesized audio via the Web Audio API.
- Tools Used: Three.js, JavaScript, WebGL, Vite

Technologies

Languages: Python, C++, C, Java, SQL, JavaScript, MATLAB

AI / ML / NLP / Vision: PyTorch, Transformers, SentenceTransformers, OpenCV, LoRA / PEFT

Robotics: ROS2, Kinematics, 3D Simulation, Three.js, Matlab

Tools & Other: Git, GitHub, Docker, Linux / Bash, REST APIs, Object-Oriented Programming, Data Structures